

COMP-SCI-5565

Intro to Statistical Learning

Topic 2 – Linear Regression

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Lecture Objectives

- To perform linear regression with a **single predictor variable**.
- To estimate the **standard error** of regression coefficients.
- To evaluate the **goodness of fit** of a regression.
- To perform linear regression with **multiple predictor variables**.
- To evaluate the **relative importance of variables** in a multiple linear regression.
- To include **interaction effects** in a multiple linear regression.
- To perform linear regression with **qualitative predictor variables**.
- To model **non-linear relationships** using polynomial regression.
- To identify **non-linearity** in a data set.

Outline

2.1 Simple Linear Regression

2.2 Assessing Accuracy of Coefficient Estimates

2.3 Multiple Linear Regression

2.4 Qualitative Predictors

2.5 Interactions & Potential Problems

Outline

2.1 Simple Linear Regression

2.2 Assessing Accuracy of Coefficient Estimates

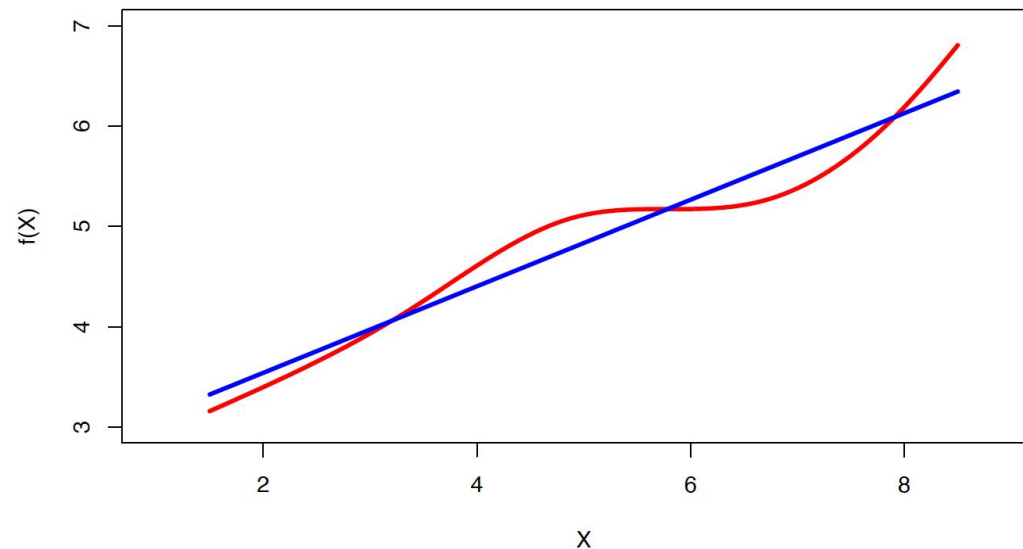
2.3 Multiple Linear Regression

2.4 Qualitative Predictors

2.5 Interactions & Potential Problems

2.1 Simple Linear Regression

- Linear regression is a simple parametric approach to supervised learning that assumes the f is linear - $y = mx + b$.
- Assumes the dependence of Y on $X_1, X_2, \dots, X_p$ is linear.

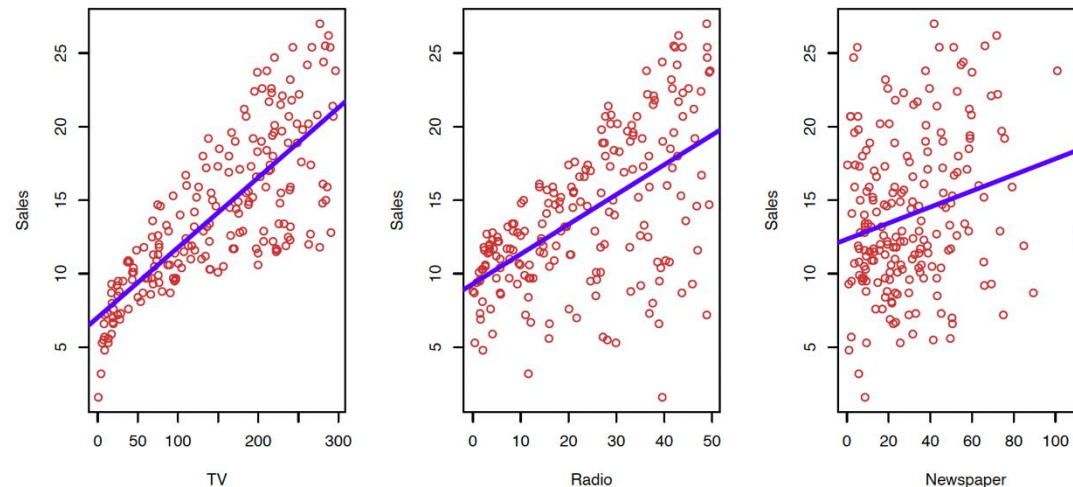


- Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

2.1 Simple Linear Regression

Questions we might ask:

- Is there a relationship between advertising budget and sales?
- How strong is the relationship between advertising budget and sales?
- Which media contribute to sales?
- How accurately can we predict future sales?
- Is the relationship linear?
- Is there *synergy* among the advertising media?



2.1 Simple Linear Regression

Simple linear regression using a single predictor X

- We assume a model

$$Y \approx \beta_0 + \beta_1 X + \epsilon,$$

where β_0 and β_1 are two unknown constants (*coefficients* or *parameters*) representing the intercept and slope, respectively, and ϵ is the irreducible error term.

- We use training data to estimate the coefficients and make future predictions, $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, where $\hat{y}$ indicates a prediction of Y based on $X = x$.
- The hat symbol indicates estimated value.

2.1 Simple Linear Regression

Estimating coefficients by Least Squares

- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, i \in \{x_i, y_i\}_1^N$
- For each i , the i^{th} residual (error) is $e_i = y_i - \hat{y}_i$
- The squared sum of all the errors – Residual Sum of Squares (RSS) can be given by

$$RSS = e_1^2 + e_2^2 + \dots + e_n^2$$

- We can rewrite RSS as

$$RSS = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \dots (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2$$

$$RSS = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- To minimize RSS, we take partial derivative of RSS with respect to $\hat{\beta}_1$ and $\hat{\beta}_0$.

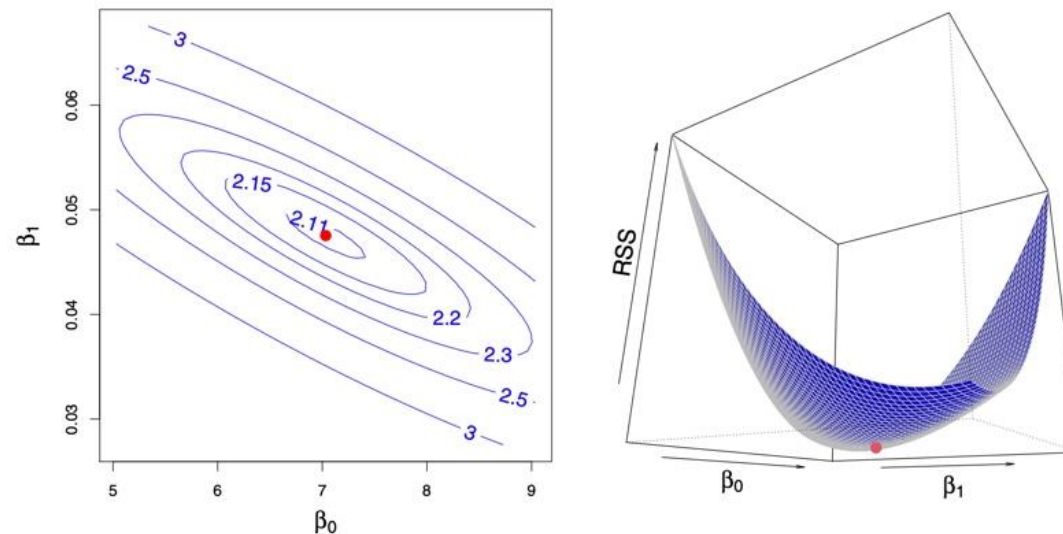
2.1 Simple Linear Regression

Estimating coefficients by Least Squares – continued

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

where $\bar{x}$ and $\bar{y}$ are samples means of x and y , respectively.



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2.2 Assessing Accuracy of Coefficient Estimates

- **Standard Error (SE):** The standard error of an estimator reflects how it varies under repeated sampling.
- We have

$$SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right],$$

$$\text{where } \sigma^2 = \text{Var}(\epsilon) = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}$$

2.2 Assessing Accuracy of Coefficient Estimates

- **95% confidence interval:** A range of values such that with 95% probability, the range will contain the true unknown value of the parameter. It has the form.

$$\hat{\beta}_1 \pm 2 \cdot SE(\hat{\beta}_1)$$

$$\hat{\beta}_0 \pm 2 \cdot SE(\hat{\beta}_0)$$

- That is, there is approximately a 95% chance that the interval

$$[\hat{\beta}_1 - 2 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot SE(\hat{\beta}_1)]$$

will contain the true value of β_1 (under a scenario where we got repeated samples like the present sample).

2.2 Assessing Accuracy of Coefficient Estimates

- **R-squared (R^2):** provides an alternative in the form of a proportion of the variance explained, ranging from 0 to 1, a good value depends on the application.

$$R^2 = 1 - \frac{RSS}{TSS}$$

where TSS is the total sum of squares:

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

2.2 Assessing Accuracy of Coefficient Estimates

Hypothesis Testing

- Standard errors can also be used to perform hypothesis tests on the coefficients – *null hypothesis* or *alternative hypothesis*.
- The most common hypothesis test involves testing the null hypothesis.
- Null hypothesis: There is no relationship between X and Y .

$$H_0: \beta_1 = 0$$

- If $\beta_1 = 0$ then the model reduces to $Y = \beta_0 + \epsilon$, and X is not associated with Y .
- Alternative hypothesis: There is relationship between X and Y .

$$H_A: \beta_1 \neq 0$$

2.2 Assessing Accuracy of Coefficient Estimates

Hypothesis Testing – continued

- To test the null hypothesis, we compute *t-statistic*, given by

$$t_{\hat{\beta}} = \frac{\hat{\beta} - \beta_0}{SE(\hat{\beta})}$$

- The t-statistic tells us how many standard-error units the coefficient is away from zero.
- Thus, if the t-statistic is greater than zero, we would reject the null hypothesis.
- It is safe to ignore the negative sign for negative values.

2.2 Assessing Accuracy of Coefficient Estimates

Hypothesis Testing – continued

- We can also use *p-value* of the t-statistic for the same purpose.
- If the p-value that corresponds to t is less than some threshold (e.g., $\alpha = 0.05$), we reject the null hypothesis.
- We conclude that there is a statistically significant relationship between the predictor variable and the response variable.

2.2 Assessing Accuracy of Coefficient Estimates

Results for the advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

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2.3 Multiple Linear Regression

- Multiple linear regression extends simple linear regression for p predictors:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon_i$$

- We interpret β_j as the average effect on Y of a one unit increase in X_j , holding all other predictors fixed.
- In the advertising example, the model becomes

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \epsilon$$

2.3 Multiple Linear Regression

Interpreting regression coefficients

- The ideal scenario is when the predictors are uncorrelated – a **balanced design**:
 - Each coefficient can be estimated and tested separately.
 - Interpretations such as “a unit change in X_j is associated with a β_j change in Y , while all the other variables stay fixed”, are possible.
- Correlations amongst predictors cause problems:
 - The variance of all coefficients tends to increase, sometimes dramatically
 - Interpretations become hazardous – when X_j changes, everything else changes.
- **Claims of causality** should be avoided for observational data.

2.3 Multiple Linear Regression

Results for advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

Correlations:				
	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

2.3 Multiple Linear Regression

Answers to the questions

- *Is there a relationship between advertising budget and sales?*
- *In other words, is at least one of the predictors $X_1, X_2, \dots, X_p$ useful in predicting the response?*
- We use **F-statistic**: A value close to 1 means there is no relationship, value greater than 1 means there is a relationship

$$F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)}$$

- An alternative is to check the t-statistic/p-values for each predictor.

2.3 Multiple Linear Regression

Answers to the questions – continue

- Which media contribute to sales?
 - p-values/t-statistic can help identify essential predictors. What if we have 200 predictors?
- How strong is the relationship between advertising budget and sales?
 - Proportion of the variance explained: R^2
- How accurately can we predict future sales?
 - Accuracy of coefficients: confidence interval
 - Test error: MSE
- Is the relationship linear?
 - Residual plots
- Is there synergy among the advertising media?
 - Interactive terms and associated p-values

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2.4 Qualitative Predictors

- Some predictors are not **quantitative** but are **qualitative**, taking a discrete set of values.
- These are also called **categorical** predictors or **factor variables**.
- Example: investigate differences in credit card balance between students and non-students, ignoring the other variables. We create a new variable

$$x_i = \begin{cases} 1 & \text{if } i \text{ th person is student} \\ 0 & \text{if } i \text{ th person is not a student} \end{cases}$$

- Resulting model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i \text{ th person is student} \\ \beta_0 + \epsilon_i & \text{if } i \text{ th person is not a student} \end{cases}$$

- Interpretation?

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Interaction/Synergy

- An *interaction term* is effectively a multiplication of two predictors you believe have a joint effect on the Y .
- Model takes the form:

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times (\text{radio} \times \text{TV}) + \epsilon$$

Results:

	Coefficient	Std. Error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

2.5 Interactions & Potential Problems

Interaction/Synergy – continue

- The results in this table suggest that interactions are essential.
- The p-value for the interaction term $TV \times radio$ is extremely low, indicating that there is strong evidence for $H_A: \beta_1 \neq 0$.
- Sometimes an interaction term has a very small p-value, but the associated main effects (in this case, TV and radio) do not.

The hierarchy principle:

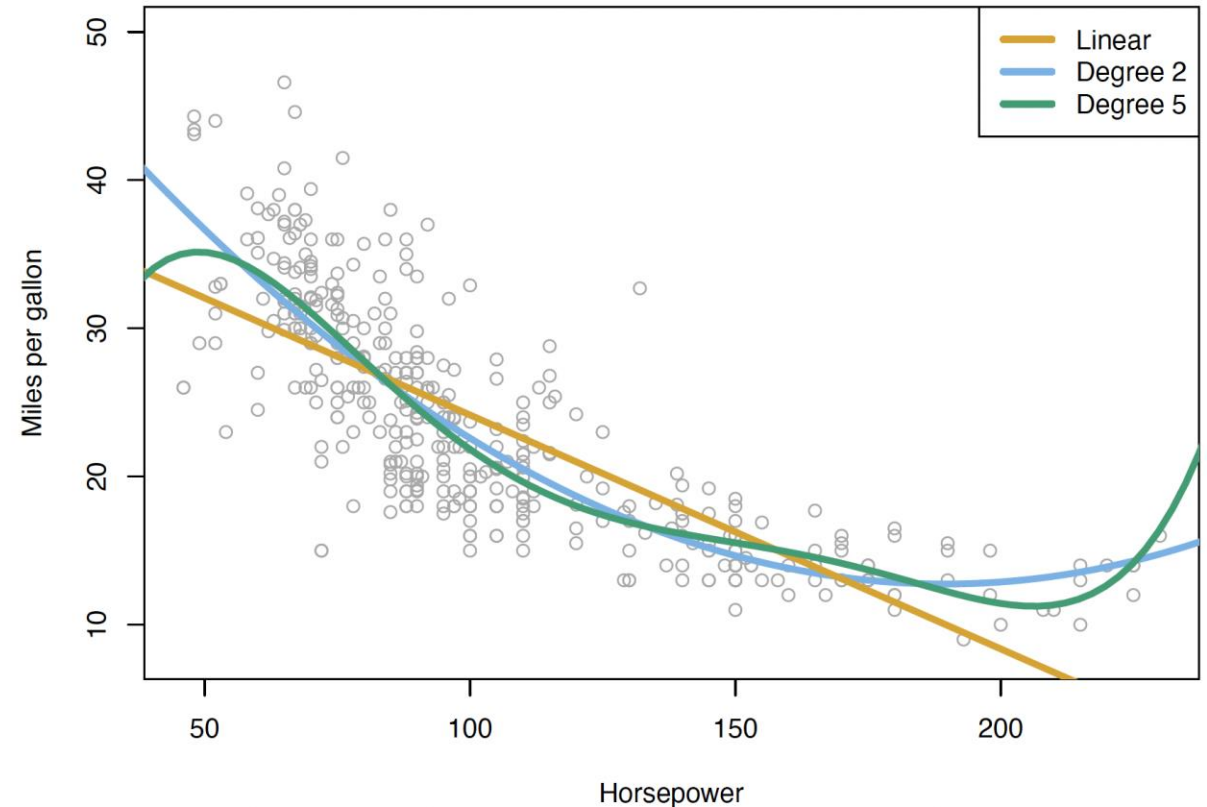
- If we include an interaction in a model, we should also include the main effects, even if the p-values associated with their coefficients are not significant.

2.5 Interactions & Potential Problems

Non-linear relationships

- In these cases, we might propose a quadratic model.
- In general, we model the expected value of y as an n^{th} degree polynomial, yielding the general polynomial regression model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots \beta_n X^n + \epsilon_i$$



2.5 Interactions & Potential Problems

Non-linear relationships – continued

- The figure suggests that

$$mpg = \beta_0 + \beta_1 \times horsepower + \beta_2 \times horsepower^2 + \epsilon$$

may provide a better fit.

	Coefficient	Std. Error	t-statistic	p-value
Intercept	56.9001	1.8004	31.6	< 0.0001
horsepower	-0.4662	0.0311	-15.0	< 0.0001
horsepower ²	0.0012	0.0001	10.1	< 0.0001

2.5 Interactions & Potential Problems

Potential Problems

- Non-linear relationships

Residual plots are helpful tools to see if any remaining trends exist. If so, consider fitting the transformation of the data.

- Outliers

An outlier may not affect the fit much but can have a dramatic effect on the RSE. Detect outliers and deal with them.

- Inability to handle categorical target variables.

Further Reading

- Chapter 3 – James G., Witten D., Hastie T., Tibshirani R., & Taylor J. (2023) *An Introduction to Statistical Learning with Applications in Python*. Springer <https://link.springer.com/book/10.1007/978-3-031-38747-0>
- Chapter 3 - Introduction to Statistical Learning Using R Book Club: <https://r4ds.github.io/bookclub-islr/>
- Summary of Chapter 3 – <https://www.bijenpatel.com/guide/islr/linear-regression/>
- Linear Regression by DeepAI – <https://deepai.org/machine-learning-glossary-and-terms/linear-regression>

End of Topic

Thank you
Any questions?